

Literature Review: ERP Implementation in Various Industries

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Abstract- An ERP system integrates all the necessary aspects of a business like purchasing, sales, product planning, finances, inventory management, through modules that share a single database. But because each company has its own diverse set of business operations and sector in which each company operates is different, no single ERP commercial package can meet the functionalities and requirements of all companies. Studies have shown that almost all ERP projects act as a challenge for any organisation in the initial phases. Therefore, the selection of an ERP process is of utmost importance for the smooth operations and profitability of any company. Also, as ERP adoption is more common these days than ever, and almost every type of industry is using these systems. In this review paper, we consider different industries and ERP implementation in respective industries.

Keywords- *ERP Implementation, ERP Benefits, ERP Applications*

I. INTRODUCTION

The main idea behind ERP was to cater for the needs of firms to manage their operations and services. Initially, ERP implementation was limited to the Manufacturing industry only and only big companies adopted this as it required huge investment and was not so popular. Many companies discovered that certain processes or department functions were not guarded, during the implementation of ERP, and they were subjected to incorporate specific application software to fill the void [1]. Gradually the popularity of ERP systems increased as they proved to be much effective in increasing the efficiency and speed of the operations in the companies, and these days majority of the industries use ERP.

However, every company has different needs and therefore needs a different strategy to implement the system.

II. LITERATURE REVIEW

In this paper, various articles about the implementation of ERP are considered and categorized industry-wise.

If we look from the manufacturing industries point of view, they were the first one to introduce and implement ERP before any other industry, as the origin of ERP is from Manufacturing Resource Planning is a derivative of Material Resource Planning, a traditional practice in most of the manufacturing firms, as stated by [2], [3], however, the term ERP was given by Gartner [4] around 1990s, and ERP implementation, due to its capability of giving an edge over the competitors, proved to be dominating the traditional systems for the planning of resources in big multinational firms [5] as the system became more flexible and could run on multiple platforms, along with integrating common manufacturing processes such as transportation, project management, financials etc [6]. Further, ERP systems became a continuous process for the companies and not just a one-time task [7].

However, ERP became one of the prime factors for industries and started to become more advanced by around the last decade [8]. Soon, as the technology and internet facilities improved drastically, ERP II started coming into picture, which performed the functions of ERP over the internet and facilitated various things apart from traditional ERP such as real time tracking of goods or raw material, enhancing e-commerce capabilities, and also improving the coordination between manufacturers and suppliers [9], and can improve the uniformity between various departments of a

company on various locations [10], thus helps to increase transparency within the organisation.

Many papers such as [11], [12] have highlighted the benefits of ERP in Manufacturing industry. The various benefits as listed by several authors include, cost reduction, increased efficiency, better monitoring etc. Also, some of the comparatively newer studies say that the benefit of the ERP systems must be clearly measurable, as it requires huge investment [13]. Some of the main benefits of ERP in manufacturing industries are reductions of errors, thus improving quality. When it comes to sales management and customer satisfaction, specially in manufacturing field, the use of ERP has assured greater customer interaction, and better understanding of customer specifications [14]. Since ERP implementation is a huge investment for the companies, measuring the benefits is also a crucial task to evaluate the performance. Though the companies claimed multiple benefits such as reducing idle time, improving financial management, etc. [15], most of the ERP systems were found to have high rate of failure initially [16].

When it comes to the health care sector, efforts to improve health care have spread over the globe, and the role of data innovation is growing to find solutions to various difficulties affecting the medical-care field. The reason for this investigation was to discover how Enterprise Resource Planning [ERP] have been utilized in the health care area and how these systems could be beneficial in improving health care management. However, other findings indicated that ERP systems might be used in the health-care sector to improve the quality of services [17]. Medical services organisations benefit and administrations could both benefit from ERP frameworks [18].

Implementation of ERP is a crucial decision for most of the companies, and in healthcare it may really mean the difference among life and death. The aggregate of the medical services industry is reliant on simple admittance to constant data, including clinical updates and charging data [19]. There are various applications inside medical care that require information, constant data updates, and information effectiveness. For instance, envision that a patient is in dire need of one specific thing, like support or a wheelchair. The staff needs to have the option to rapidly check their stock and check whether it is there in supply [20]. If not, they should have the option to rapidly arrange more to guarantee the patient gets the facilities they need.

Data is a significant part of the medical services industry. Sharing information has gotten significant to decrease costs and improve the administrations delivered to clients. It additionally needs to give data about charging and clinical information inside the medical services association and to partners [21]. In addition, its expanded utilization has been driven by the craving to diminish costs, to increment competitiveness in the medical care area, and to improve customized individual medical care [22].

The health care industry is firmly relying upon its clients and data identifying with the customers, drug stores and other specialist co-ops [23]. Looking for approaches to improve service quality is identified with costs and more popularity in the administrations [24]. As a result of competition among medical care suppliers, medical services organisations have changed quickly over the most recent couple of years in order to improve the adequacy and efficiency of medical care conveyance [25]. ERP frameworks have been known in assembling and administration organisations since 1990's and later mid 2000 they have been executed likewise in medical care area [26].

ERP frameworks are programming bundles that permit an association to see data about the whole association, just as the ability to impact the tasks of an association. [27] characterizes ERP as 'programs that give single coordinated programming that expect to give single incorporated delicate product to deal with various corporate capacities including account, HR, fabricating, materials the board etc.... Clinicians, doctors, attendants and directors need quality instruments to improve and adapt to arising clinical advancements and strategies to keep up quality medical care administrations.

Medical care associations handle gigantic measures of data that they depend on to complete their services [28]. The data is utilized variety as per the capacity of offices, like controlling and arranging. Subsequently, it is important to make this data available to the fitting offices at the ideal time [29].

Coordinated frameworks are a significant part of each help association; the medical services area is no special case [30] and integrated electronic data frameworks can successfully change the medical care area [31]. For instance, the utilization of telemedicine is expanding, taking into consideration the organization of medical services to far off patients [18].

Despite the fact that ERP frameworks are prescribed as an answer for a few difficulties, [32] guaranteed that these frameworks are exorbitant and that they change the activity of an association. Associations that have effectively actualized an incorporated an ERP framework may confront opposition from representative clients that are needed to move to the new frameworks. In view of the protection from accepting change, it thusly could take a long effort for the association to understand the advantages of the new framework, which may bring about spending invades.

In recent years, educational institutions have been evolving and becoming updated with technology. Whether we consider schools or higher educational institutions, every institute needs to manage large number of students and staff data, along with all the schedules and academic resources. As a result, almost all the educational institutions are implementing computer systems for better efficiency in management and administration [33], [34]. In education industry, the aim of investing in a project or implementing ERP is to enhance research and teaching for reasonable costs [35]. However, many studies have also the opposite results claiming that many ERP systems fail to meet the expectations. In case of educational institutes, they differ from other organisations or industries as they often use ERP systems for academic purposes and managing student databases [36]. An ERP system is found to facilitate any institution's mission by providing better communication and interaction between faculty and students and provide better E-learning environment. Though, like other industries, measuring success of ERP implementation is still a challenge, keeping in mind that implementation requires huge investment. So, studying the impact of ERP implementation on user performance is an effective way of measure the output of ERP.

During the initial phase of ERP implementation in education industry, [14] analysed the implementation of ERP in Swiss university, during that time in 1999, when ERP implementation in university was not common, and found that these systems will be definitely more effective in future due to better interactive facilities between different faculties, students and staff. After that, [37] conducted a study which focussed on trends in adopting ERP systems for universities, however, the study was only based upon the secondary data from the internet sources. Other researches, such as [38]–[40] have shown universities want a more efficient system but the results were limited and staff also needed to put effort to adopt new ways.

In case of educational institutes, investment in ERP systems, like any other industry is huge, but on the other hand, their profit is often not immediate and in most of the cases, it

is long term [41]. Several other researches, also focused on the difficulties other than investment, like adding all the previous databases which were many years old, to the system would require additional team of employees may result in decreased efficiency [34], [42]–[44].

The hospitality industry has to offer flexibility and performance, permitting to meet customers needs [45]. The ERP systems contain modules that merge records into one database and one interface. But every industry has their own unique needs. Hence, specified modules were developed by ERP carriers, integrated into the system, for specified actions in the industry [46].

Data integration is witnessed as the main problem in hospitality management, making it a challenging task to produce reports with aggregate information [46]. In order to solve the problem of not having a single integrated solution, integrated tools were developed by the vendors of ERP software [47]. With respect to the hospitality industry, many of the vendors of ERP have focused on their commercial strategies to cover back office operations, that are typically the same regardless of the industry we consider. For example, an international hotel chain ERP has adopted systems that were habitually owned in some other industries as well. SAP exists as a market leader in terms of hospitality industry and other industries as well when it comes to ERP systems. Industry-specific operations, such as front office, reservation, inventory, food and beverage or POS (points of sale) are noticed by specific solutions, and are not often integrated with back office and system ERP executed operations [46].

The hospitality industry has undergone significant transformation in recent years, with information technology playing a critical role. There's a need to make use of newer technologies to "re-invent" both the external processes and internal processes to boost performance and improve customer loyalty [48]. Other authors emphasise the need for the company to adapt quickly to new customer demands in terms of service quality, suggesting a strategic use of IS/IT [49], and others [47]. Another critical success factor [CSF] of this company is its partnership with external operators across new distribution networks, which could result in significant benefits utilising IS/IT [48]. Another CSF in the hospitality industry is diversifying the service offering. Services such as entertainment, health centres, and golf courses are listed as differentiators against competitors by this author. This corroborated what authors such as [50] had said about the value of service offer diversification as a differentiator. Organizations must constantly assess if their operations are in sync with the external environment with the aim of maintaining improvements in necessary and, where possible,

integrated IS/IT structures. The investments in IS/IT must be handled, efficiently and effortlessly for medium and long-term market sustainability [51]. Just 30% of IS/IT investment projects attain the intended benefits, according to some authors [52], which is one justification for using a reference model of best practices to help mitigate this danger.

ERP systems were developed as management systems to allow integrated resource management by automating many of the tasks or processes across multiple sections of an enterprise in order to process data in real time. This definition suggests a holistic view of all organisational units, as well as frameworks directed to the enterprise as a whole. This was the theoretical viewpoint the investigation got to know on consulted writers [53], in the literature review [51]. It was also discovered that, considering all of the advantages obtained through the introduction of these programmes, they do not cover all processes in all industries.

Another CSF, process performance enhancement, was found to be harmed by the lack of coordination between the front office system and tourism advisors such as Booking.com, Ebix etc.. Due to the lack of integration, the effectiveness of these systems is calculated manually, by human hand, using conventional methods such as fax or e-mail. As a result, aside from inefficiency and time duplication in completing the tasks essential in the processes, the hotel management failed to capitalise on one significant ability of IS/IT in this industry: integrating the value chain processes into an integrated information system, as highlighted by authors such as [50].

Other researchers [49], [47] found similar findings in their investigations. Poor data integration limits an organization's capacity to interact with other business partners because inter-organizational integration is lacking, and therefore the internet and other technologies' capacities are not being utilised.

According to [46], many ERP vendors have focused their attention on back office processes to understand how ERP systems can add value to the hospitality industry. These processes are very similar in this industry to those in other industries. Large hotel units use ERP systems that are commonly used in other industries.

III. CONCLUSION/ FUTURE RESEARCH

In the paper, articles about implementation of ERP in specific industries were reviewed and it can be seen that ERP implementation is an essential practice in most of the companies or industries today, but for each company, it has different objectives, strategies, history and challenges.

In Manufacturing industry, which is the origin of ERP, the system mainly focusses on tracking of materials, production scheduling, resource management etc. most of the firms experienced an edge over the competitiveness due to the increased efficiency in material planning, tracking, keeping up with the schedules, transportation, and improving the coordination between different departments. Though the ERP systems failed initially, they see a huge rise afterwards in other industries as well.

In case of Education industry, ERP plays a completely different role as compared with other industries, i.e. it is seen as an asset used for management of academic and research facilities, along with managing the staff and student data. Today, almost all higher education institutes use ERP, though initially it was doubted as huge investment is needed and educational institutions have comparatively lesser profits than big companies.

In the case of healthcare industry, ERP has significant role of managing the essential supplies, tracking the facilities, and assisting the administration. In most of the big hospitals today, all the patient's records are integrated into a common portal, thus drastically reducing the risk of losing the information and makes easier for both the patients and doctors to keep a track of the treatment. Apart from that, ERP systems also play a huge role in managing the crowd at the hospitals or other healthcare facilities. If we look from medical suppliers point of view, it increases competitiveness for them as it has resulted in faster and more efficient conveyance.

Coming to the hospitality industry, ERP is an essential requirement for booking of rooms, monitoring the staff, scheduling the processes such as cleaning, maintenance etc. and taking care of other activities inside the hotels. It is also seen that it plays a crucial role in maintaining customer loyalty and an edge over the competitors as many entertainment and luxurious facilities are smoothened by ERP systems. One of the main changes that hospitality industry has witnessed due to the implementation of ERP is that they take online bookings from third party tourism websites and integrate it with their booking portal, thus ensuring much more efficient bookings procedure for both customers and the hotels.

As seen from the literature review considering the ERP implementation in various industries, it is a promising field of research in coming years because of its wide applicability and necessity across the world in almost all the industries. ERP implementation will be seen as a tool for competitive advantage, always having a scope for improvement. One of the major gaps is still the heavy investment required for the implementation of ERP systems. Though there is significant research in measuring the applicability of ERP systems in the industries, it is only seen as a hefty and careful investment. There is very limited research on how to make these systems more standardized and affordable.

Also, the majority of the publications have considered the effectiveness of ERP systems but there have been limited number of the researchers who have focussed on the post implementation phase of ERP systems. As already highlighted in the paper, ERP implementations are subject to huge investments and failure may lead to significant losses, there may be increasing trend of research in the post implementation phase which may highlight or suggest certain standardized measures to take the utmost benefit from already implemented systems. This will add to the resilience in ERP implementation, making it a safer option and reducing the risk factor, so that even the small organisations are more confident in trying ERP implementation.

IV. REFERENCES

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